

* Compare the Instruction Set of Thumb and Thumb-2 Technology :-

Thumb

Classic ARM processor: Ex ARM7TDMI

It includes all high performance Instructions

ARM Instruction Set:
32 bit Instructions
↳ For higher performance

It includes 16-bit Instructions. provides excellent code density. It doesn't provide high performance Instructions.

More Instructions are needed to perform high operations.

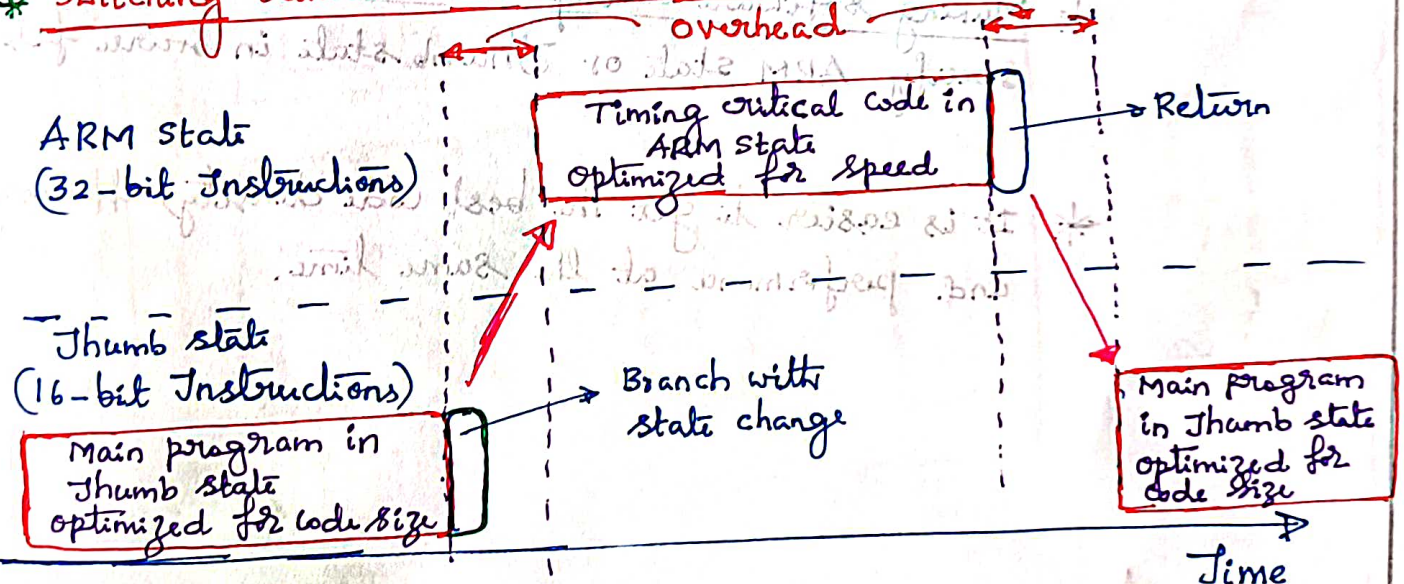
switching b/n ARM state and Thumb state

Cortex-M processors

Thumb Instruction Set
(16-bit & 32-bit Instructions)

No state switching is required

* switching between ARM code and Thumb code in classic ARM processors.



- ↳ For many applications using classic ARM processors, switching state is required to get the benefits of ARM and Thumb states.
- ↳ As shown in the above fig, there is overhead (in terms of both execution time and Instruction count) involved while switching b/n the states.

→ The separation of two states increases the complexity of software compilation process and its harder to develop the optimized code.

→ Thumb-2 Technology:

→ In this Technology, Thumb Instruction set has been extended to support both 16-bit and 32-bit Instruction encoding.

→ Since it has only one state, no switching between the states is required.

→ With Thumb-2 technology, the Cortex-M processor has a number of advantages over classic ARM processors

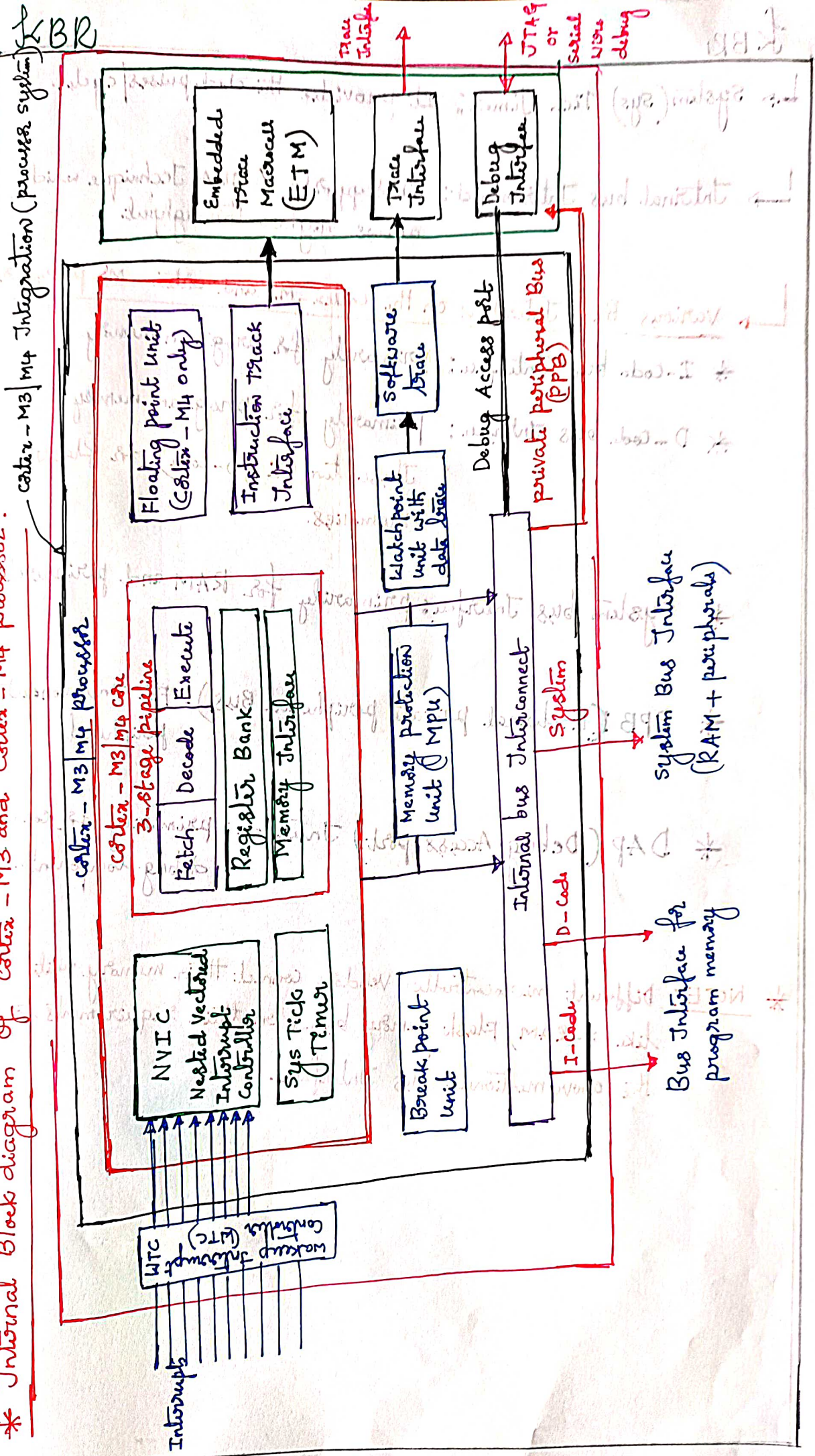
→ Advantages:

* No state switching overhead: saving switching time, execution time and Instruction space.

* Making software development easier: No need to specify ARM state or Thumb state in source file

* It is easier to get the best code density, efficiency and performance at the same time.

* Internal Block diagram of cortex - M3 and cortex - M4 processor:



LBR

↳ System(sy) Tick Timer: It provides the clock pulses/cycles.

↳ Internal bus Interconnect: It supports AMBA Technique which allows higher throughput.

↳ Various Bus Interfaces on the cortex-M3 and cortex-M4 Processors

* I-Code bus Interface: primarily for program memory

* D-Code bus Interface: primarily for program memory

: These two (I & D-code) for flash memories.

* System bus Interface: primarily for RAM and peripherals

* PPB (External private peripheral BUS): For interconnecting peripherals

* DAP (Debug Access port) Interface: primarily for connecting debug components.

* NOTE: Different microcontroller vendors connect their memory units like S-RAM, Flash memory based on their requirements to the above mentioned bus Interfaces.